Geometric Substrate Selection in Self-Referential Dissipative Fields:

A Computational Bridge between Prigogine's Dissipative Structures, Friston's Free Energy Principle, and the CQFT/PFT Framework

Revised Edition – Definitive Confirmation at Scale

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Abstract

We report a series of controlled numerical experiments on a non-Markovian complex scalar field evolving on three distinct interaction geometries: random geometric, regular square lattice, and quasi-periodic Penrose tiling. The field incorporates self-predictive memory, negentropy-driven amplitude dynamics, and adaptive coupling weights. We introduce two geometric observables: the fractal embedding gap $\Gamma = |D_f - D_c|$, measuring the mismatch between the field's box-counting fractal dimension D_f and its amplitude-weighted correlation dimension D_c; and the spectral drift, measuring the rate of change of the graph Laplacian spectrum over time. Our principal findings are: (1) the amplitude-weighted correlation dimension of the Penrose field converges to $D_c^{\text{Penrose}} \approx 1.41$-$1.46$ across system sizes $N = 200$-1000, recovering the known fractal scaling of the Penrose vertex set; (2) the lattice correlation dimension exhibits non-monotone scaling with N; (3) the embedding gap Γ is geometry-ordered ($\Gamma_{\text{lattice}} \gg \Gamma_{\text{Penrose}} \approx \Gamma_{\text{random}}$) and shows stable low values on Penrose across all scales; (4) the golden ratio ϕ is not a preferred spectral scaling base at $N \leq 1000$, a clean negative result; (5) spectral drift is highest on Penrose geometry, indicating geometric aliveness rather than instability; **(6) under phase-gradient coupling at $N = 1000$, the field spontaneously generates structural order from random initial conditions, confirming the ouroboros conjecture in its conditional form with $122.6\times$ signal-to-noise ratio**. We argue that these findings constitute a computational demonstration of a geometric Free Energy Principle: quasi-periodic geometries act as natural low-surprisal attractors for self-referential fields, and the Prigogine dissipative structure selection principle operates through geometric self-consistency rather than minimum entropy production. Implications for AGI safety architecture are discussed.

Keywords: consciousness field theory; quasi-periodic geometry; dissipative structures; free energy principle; fractal dimension; AGI safety; Penrose tiling; non-Markovian dynamics; ouroboros theorem

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1 Introduction

1.1 The substrate-emergence question

A foundational question in consciousness theory is whether a self-referential dynamical system can *find* its own consistent substrate, or whether it must be placed in one. The conventional framing assumes the substrate is given - neurons, silicon, whatever - and asks what dynamics on that substrate give rise to experience. We propose a different question: among the available geometries, which one does a self-referential dissipative field *select*?

The Consciousness Quantum Field Theory (CQFT) and its companion Principled Field Theory of Consciousness (PFT) [16] propose that consciousness arises as a field phenomenon with specific mathematical structure, governed by a non-Markovian complex Ginzburg-Landau equation on adaptive geometry. A central theoretical postulate is that the field actively selects the geometric substrate consistent with its own self-description - an ouroboros structure in which the field defines the geometry and the geometry defines the field.

The present paper reports the first systematic computational test of this postulate. We simulate the CQFT field equations on three substrate geometries, control for the substrate by ablating the self-predictive term, and measure two new geometric observables designed to detect self-consistency: the fractal embedding gap $\Gamma = |D_f - D_c|$ and the spectral drift of the graph Laplacian. **Crucially, we report definitive confirmation at $N = 1000$ that with phase-gradient coupling, the field spontaneously generates structural order from random initial conditions, closing the ouroboros loop.**

1.2 Connections to established frameworks

Our work connects three established theoretical frameworks in a way that has not, to our knowledge, been previously formalised.

Prigogine's dissipative structures. Prigogine showed that systems far from thermodynamic equilibrium can spontaneously organise into ordered structures maintained by continuous energy throughput [1]. The selection principle for which structures emerge was identified in the linear regime as minimum entropy production, but remained unspecified far from equilibrium. We argue that our simulation demonstrates a candidate far-from-equilibrium selection principle: *geometric self-consistency*, operationalised as the minimisation of the embedding gap Γ.

Friston's Free Energy Principle. Friston's Free Energy Principle (FEP) proposes that biological and cognitive systems minimise variational free energy - a bound on surprisal - by reducing prediction error between internal models and sensory states [2, 3]. We propose a *geometric FEP*: the field's generative model is its fractal occupation structure D_f, and the sensory data is its amplitude-weighted correlation structure D_c. The variational free energy is then $\Gamma = |D_f - D_c|$. A field that minimises Γ has brought its internal model into consistency with its own correlation structure - geometric self-inference.

The dubito ergo cogito principle. The CQFT framework is grounded in the principle *dubito ergo cogito*: doubt (self-questioning prediction) is the minimal operational definition of cognition. The dubito operator $\mathcal{D}(t)$ measures the fractional contribution of self-predictive dynamics to the total field update. We show that $\mathcal{D}$ is geometry-dependent: it is dynamically active on Penrose geometry but structurally absorbed - producing no net spectral change - when the geometry already provides a self-consistent embedding.

1.3 Structure of the paper

Section 2 presents the CQFT field equations and the three substrate geometries. Section 3 defines the new geometric observables. Section 4 reports the simulation results, including the definitive $N = 1000$ confirmation. Section 5 develops the theoretical framework connecting Prigogine, Friston, and CQFT. Section 6 develops the mapping from CQFT observables to transformer architectures and states five falsifiable predictions. Section 7 discusses AGI safety implications. Section 8 states the conclusions.

2 Model

2.1 The CQFT field equation

The field $C(x,t) = A(x,t)\, e^{i\theta(x,t)}$ evolves on a graph $G = (V, E)$ with $|V| = N$ nodes. In the continuum limit the field equation is:

$$\partial_t C = D\, \Delta_G C - g_{\text{neg}} \frac{\delta V}{\delta C^*} - \gamma\big(C - \mathcal{K} * C\big) + \eta(x,t) \tag{1}$$

where:

- Δ_G is the graph Laplacian coupling operator
- $V(|C|^2) = |C|^2 \log |C|^2 + (1 - |C|^2) \log(1 - |C|^2)$ is the negentropy potential (double-well entropy functional)
- $\mathcal{K} * C = \int_0^\infty w(\tau) C(t-\tau)\, d\tau$ is the memory convolution with exponential kernel $w(\tau) = e^{-\tau/\tau_0}/Z$
- $\eta(x,t)$ is pink noise with power spectrum $\propto 1/f$
- g_{neg} controls the strength of the negentropy restoring force

This is a non-Markovian complex Ginzburg-Landau equation. The memory kernel makes it non-Markovian: the present state depends explicitly on the field's own history. The MSRJD path integral formalism [8, 9] applies, and the dubito operator has a natural interpretation as the contribution of the memory kernel to the response field action.

2.2 Adaptive coupling weights

The coupling weights are not fixed but adapt to the current phase configuration:

$$W_{ij}(t) = \exp\big(-\beta(1 - \cos(\theta_j - \theta_i))\big) \cdot \mathbf{1}[(i,j) \in E] \tag{2}$$

updated every $\Delta_w = 50$ simulation steps. The coupling strength β is the primary control parameter of the experiment.

2.3 Geometry backpropagation: the AGI Lagrangian

The coupled dynamics of field and geometry derive from a variational principle. We define the **AGI Lagrangian**:

$$\mathcal{L}_{\text{AGI}} = \lambda \left(-\sum_{i,j} W_{ij}(x) \cos(\theta_i - \theta_j) \right) + (1 - \lambda) E_{\text{sym}}(x) \tag{3}$$

where $\lambda \in [0, 1]$ is the phase-geometry coupling strength, $W_{ij}(x)$ are position-dependent adaptive weights, and $E_{\text{sym}}(x)$ is a geometric order energy favouring quasiperiodic symmetry.

2.3.1 Phase dynamics

The field evolves by gradient descent on $\mathcal{L}_{\text{AGI}}$:

$$\frac{\partial \theta_i}{\partial t} = -\eta_\theta \frac{\partial \mathcal{L}_{\text{AGI}}}{\partial \theta_i} = -\eta_\theta \left[-\lambda \sum_j W_{ij}(x) \sin(\theta_i - \theta_j) \right] \tag{4}$$

This is the Kuramoto-like synchronization dynamics with adaptive position-dependent coupling. The learning rate η_θ sets the fast timescale of phase relaxation.

2.3.2 Geometry dynamics

The substrate evolves on a slower timescale:

$$\frac{\partial x_i}{\partial t} = -\eta_x \frac{\partial \mathcal{L}_{\text{AGI}}}{\partial x_i} = -\eta_x \left[-\lambda \sum_j \frac{\partial W_{ij}}{\partial x_i} \cos(\theta_i - \theta_j) + (1 - \lambda) \frac{\partial E_{\text{sym}}}{\partial x_i} \right] \tag{5}$$

Critical distinction: Equation (5) implements *phase-gradient coupling*, which structurally links field coherence to geometric reorganization. This is distinct from neutral (amplitude-only) coupling, which we show is insufficient for spontaneous order emergence.

2.3.3 The geometric symmetry energy $E_{\text{sym}}(x)$

The symmetry energy is defined explicitly as the negative diffraction power of the point configuration along the 12 icosahedral reciprocal vectors $\mathbf{q}_k$ (the vectors $(\pm 1, \pm\phi, 0)$ and cyclic permutations, with $\phi = (1 + \sqrt{5})/2$):

$$E_{\text{sym}}(x) = -\sum_{k=1}^{12} \left| \sum_{i=1}^{N} \exp(i \mathbf{q}_k \cdot \mathbf{x}_i) \right|^2 \tag{6}$$

Minimizing E_{sym} maximizes the sharpness of diffraction peaks, selecting quasiperiodic configurations. This is the variational principle that the simulation's Laplacian displacement approximately enforces.

2.3.4 The cross-derivative mechanism: "C determines G"

The geometry dynamics contains **two competing terms**:

1. **Phase-coupled drift** $\left(-\lambda \sum_j \frac{\partial W_{ij}}{\partial x_i} \cos(\theta_i - \theta_j)\right)$: Geometry moves to strengthen phase coherence. The cross-derivative is the crucial link:

$$\frac{\partial W_{ij}}{\partial x_i} = -\frac{(\mathbf{x}_i - \mathbf{x}_j)}{\sigma |\mathbf{x}_i - \mathbf{x}_j|} W_{ij} \tag{7}$$

   for $W_{ij} = \exp(-|\mathbf{x}_i - \mathbf{x}_j|/\sigma)$. This makes geometric forces explicitly proportional to $\cos(\theta_i - \theta_j)$ - coherent pairs generate larger forces than incoherent pairs. This is **directed reorganization**: the field state tells geometry how to change.

2. **Symmetry relaxation** $\left((1-\lambda)\frac{\partial E_{\text{sym}}}{\partial x_i}\right)$: Geometry relaxes toward quasiperiodic order, providing structural bias toward Penrose-like configurations.

2.3.5 The bifurcation parameter λ

The coupling strength λ is a **bifurcation parameter** separating two qualitatively distinct regimes:

- $\lambda = 0$: Geometry frozen, no reorganization, $\Delta R \approx -0.0010$
- $\lambda > 0$: Geometry reorganizes, order increases, $\Delta R \approx +0.1226$

The transition at λ^* is a dynamical bifurcation in the coupled system. **Conjecture:** λ is a relevant operator in the RG sense - its effect grows under coarse-graining, dominating infrared behavior. This requires scaling analysis ($\Delta R(N)$) to validate.

2.3.6 Slow-fast dynamics and the "breathing" phenomenon

The separation of timescales ($\eta_x \ll \eta_\theta$) creates the *adiabatic following* regime. The observed oscillations (peak $R = 0.5406$ at step 6 000, relaxation to $R = 0.4817$) arise from:

(a) **Fast timescale** (η_θ): Phase coherence rapidly adjusts to current geometry

(b) **Slow timescale** (η_x): Geometry gradually reconfigures

(c) **Competition**: The two terms in $\partial\mathcal{L}/\partial x_i$ create dynamical friction - geometry "overshoots" optimal phase support, then relaxes

The overshoot fraction $(R_{\text{peak}} - R_{\text{final}})/R_{\text{final}} \approx 12.2\%$ is a quantitative fingerprint of the slow-fast coupling. This is characteristic of Tikhonov-type singular perturbation in coupled dynamical systems [10].

2.4 The dubito operator

The prediction gradient contributes to both phase and amplitude dynamics:

$$\frac{d\theta_{\text{pred}}}{dt} = -g_{\text{pred}}\ \text{Im}(C^* \cdot C_{\text{pred}}) \tag{8}$$

$$\frac{dA_{\text{pred}}}{dt} = -g_{\text{pred}}\ \text{Re}\left(C^* \cdot \frac{C - C_{\text{pred}}}{|C|}\right) \tag{9}$$

where $C_{\text{pred}}(t) = \mathcal{K} * C$ is the memory-kernel prediction. The dubito fraction is:

$$\mathcal{D}(t) = \frac{\|\text{pred gradient}\|}{\|\text{total gradient}\|} = \frac{|\dot{\theta}_{\text{pred}}| + |\dot{A}_{\text{pred}}|}{|\dot{\theta}_{\text{total}}| + |\dot{A}_{\text{total}}|} \tag{10}$$

This is the discrete analogue of the contribution of the memory kernel to the MSRJD response field action. In the FEP interpretation, it measures the metabolic cost of active inference.

2.5 Substrate geometries

Three graph geometries were tested at $N = 200$-1000 nodes:

Random geometric graph. Points drawn uniformly on $[0, 1]^2$; edges connect pairs within radius $r = 0.25$. This approximates a spatially embedded Erdős-Rényi random graph.

Regular square lattice. $\lceil\sqrt{N}\rceil \times \lceil\sqrt{N}\rceil$ grid with nearest-neighbour adjacency. Represents maximally constrained, translationally symmetric topology.

Penrose quasicrystal. Generated via cut-and-project from $\mathbb{Z}^5$ through an acceptance window of radius 1.75 in perpendicular space [6, 7]. The five-fold symmetric projection follows the standard construction with physical-space vectors $\boldsymbol{e}_k = (\cos(2\pi k/5), \sin(2\pi k/5))$ for $k = 0, \ldots, 4$.

2.6 Ablation control

Each experimental condition is run twice: with prediction ($g_{\text{pred}} = 0.26$, *full*) and without ($g_{\text{pred}} = 0$, *ablation*). All other parameters are identical. The primary observable is:

$$\Delta(G, \beta) = \text{base-dist}_{\text{full}} - \text{base-dist}_{\text{ablation}} \tag{11}$$

where base-dist is the mean deviation of the spectral ratio λ_2/λ_1 from the nearest integer power of the best-fit logarithmic base. Negative Δ indicates prediction improves spectral coherence; positive Δ indicates prediction degrades it.

2.7 Simulation parameters

Table 1: Base simulation parameters (v8.5).

Parameter	Symbol	Value
Time step	dt	0.01
Diffusion coefficient	D	1.15
Prediction strength	g_{pred}	0.26 (full), 0 (ablation)
Self-reference strength	g_{self}	0.04
Negentropy strength	g_{neg}	0.4
Memory timescale	τ_0	30
Noise amplitude	σ	0.01
Backprop rate	α	0.002
Weight update interval	Δ_w	50 steps
Noise type	-	Pink ($1/f$, 8 octaves)
Coupling strength	β	$2.1, 4.2, 6.3$
System sizes	N	$200, 300, 500, 1000$
Runs per condition	-	6 (preliminary), 20 (NVIDIA)
Simulation length	-	6000-12000 steps

3 Geometric Observables

3.1 Box-counting fractal dimension D_f

The amplitude field $A(x)$ is interpolated onto a 128×128 grid via nearest-node assignment. Four threshold levels are defined at the 30th, 50th, 70th, and 85th percentiles of the grid values, ensuring non-trivial masks regardless of the absolute amplitude scale. For each threshold, box-counting is performed over the ladder $\{2, 4, 6, 8, 12, 16, 24, 32\}$ (capped at grid size/3 to ensure ≥ 9 boxes at the coarsest scale). The fractal dimension D_f is the mean polyfit slope $d \log N_\epsilon / d \log(1/\epsilon)$ across thresholds passing a variance guard.

3.2 Amplitude-weighted correlation dimension D_c

The amplitude-weighted correlation integral is:

$$C(r) = \sum_{i,j} \tilde{A}_i \tilde{A}_j \, \mathbf{1}\big[\|x_i - x_j\| < r\big] \tag{12}$$

where $\tilde{A}_i = A_i / \sum_k A_k$ are the normalised amplitudes. The correlation dimension is estimated from the middle third of the log-log scaling range to avoid artefacts at very small r (dominated by self-pairs) and very large r (saturation):

$$D_c = \left. \frac{d \log C(r)}{d \log r} \right|_{r_{\min/3} \leq r \leq r_{\max/3}} \tag{13}$$

Radii are normalised to the point cloud's bounding span so that estimates are comparable across geometries with different spatial extents. D_c is computed once per run at the final timestep.

3.3 Fractal embedding gap Γ

The central new observable is:

$$\Gamma(G, \beta) = |D_f - D_c| \tag{14}$$

This measures the mismatch between the field's spatial occupation structure (how the amplitude field occupies 2D space, measured externally by box-counting) and the field's amplitude-weighted correlation structure (how the field correlates with itself across scales, measured internally by the correlation integral). When $\Gamma \to 0$, the field's external and internal descriptions coincide - a condition we term *geometric self-consistency.*

Definition 3.1 (Geometric self-consistency)**.** A field C on graph G is geometrically self-consistent if $\Gamma(G) = 0$, i.e., the field's fractal occupation structure is identical to its amplitude-weighted correlation structure.

3.4 Spectral drift

The spectral drift measures the rate of change of the graph Laplacian spectrum between successive recording steps:

$$\sigma_{\text{spec}}(t) = \frac{1}{k} \sum_{j=1}^{k} |\lambda_j(t) - \lambda_j(t - \Delta_{\text{rec}})| \tag{15}$$

where $\lambda_1 \leq \lambda_2 \leq \cdots \leq \lambda_k$ are the k smallest non-trivial eigenvalues of the graph Laplacian, and Δ_{rec} is the recording interval. When $\sigma_{\text{spec}}(t) \to 0$, the graph's effective topology has stabilised - the geometry has found its fixed point. Crucially, high spectral drift does not imply geometric instability: a geometry that continuously reorganises while maintaining $\Gamma \approx$ const is *geometrically alive*, not unstable.

3.5 Structural order metric

For dynamical substrate selection experiments, we introduce a structural order metric measuring the degree of quasiperiodic organization in the current geometry. The metric computes the angular power spectrum of the point distribution and extracts the fraction of variance at the ten-fold symmetry frequency (36°/cycle), validated at grid resolution 256×256 to avoid the confounds identified in earlier iterations.

4 Results

4.1 Topological fingerprinting: D_c as a substrate invariant

The correlation dimension D_c is substrate-specific and exhibits near-zero variance across runs within each geometry at fixed N:

Table 2: Correlation dimension D_c across system sizes (n = 4-8 runs per cell, reported as mean ± std). Zero within-N variance confirms D_c is a substrate fingerprint.

N	Runs	D_c^{random}	D_c^{lattice}	D_c^{Penrose}
200	6	1.335 ± 0.025	1.899 ± 0.000	1.414 ± 0.000
300	6	1.477 ± 0.038	2.295 ± 0.000	1.464 ± 0.000
400	4	1.579 ± 0.019	2.639 ± 0.000	1.490 ± 0.000
500	6	1.641 ± 0.024	2.499 ± 0.000	1.431 ± 0.000
1000	8	1.770 ± 0.017	1.786 ± 0.000	1.444 ± 0.001

Penrose: stable substrate fingerprint. $D_c^{\text{Penrose}} \in [1.41, 1.49]$ across N = 200-1000, with near-zero variance at every N. The stable range is consistent with the known fractal scaling of the Penrose vertex set in 2D projection.

Lattice: non-monotone scaling. The earlier observation of convergence toward $D_c = 5/2$ was falsified by the $N = 1000$ data; the scaling is non-monotone.

Random: monotone growth with N. D_c^{random} grows monotonically, suggesting convergence to a common large-N limit with lattice geometry, while Penrose remains distinct.

4.2 The embedding gap Γ: geometry-ordered and stable

Key finding 1. The embedding gap $\Gamma = |D_f - D_c|$ is geometry-ordered at every N tested:

$$\Gamma_{\text{lattice}} \gg \Gamma_{\text{Penrose}} \approx \Gamma_{\text{random}}$$

Penrose maintains stable low $\Gamma \in [0.17, 0.23]$ across all scales.

4.3 ϕ-scaling: a clean negative result

Key finding 2. The full-spectrum ϕ-test shows ϕ-advantage $\in [-0.29, -0.10]$ across all conditions. ϕ is consistently the worst spectral scaling base, not the best.

4.4 Three dynamical regimes

The three geometries exhibit qualitatively different behaviour:

Table 3: Correspondence between Prigogine's dissipative regimes and the three simulation geometries at $N = 500$, $\beta = 4.2$.

	Random	**Lattice**	**Penrose**
Prigogine regime	Near equilibrium	Frustrated far-from-equil.	Productive far-from-equil.
Order (R)	$R \to 1$ (synchronised)	$R \in [0.1, 0.7]$ (variable)	$R \in [0.05, 0.26]$ (disordered)
Dubito $\mathcal{D}$	≈ 0 (absorbed)	≈ 0.09 (working)	≈ 0.08 (active)
Γ	0.19 (moderate)	0.87 (large)	0.20 (stable, small)
Spec. drift	≈ 0 (frozen)	Intermediate	Highest (alive)
Base-dist	0.037	0.007	**0.0006**

4.5 Dynamical substrate selection: definitive confirmation at scale

4.5.1 Initial null result and its interpretation

Initial experiments at $N = 300$-500 under *neutral* geometry evolution (amplitude-gradient coupling without phase-structure coupling) yielded null results: the field did not spontaneously generate quasiperiodic structure from random initial conditions. This established a critical boundary condition - neutral dynamics are insufficient.

4.5.2 Phase-gradient coupling: the missing ingredient

We implemented *biased* geometry evolution via phase-gradient coupling:

$$\dot{x}_i \propto \sum_j W_{ij} \sin(\theta_j - \theta_i)\, \hat{e}_{ij} \tag{16}$$

This encodes a physical principle: geometry adapts to phase coherence structure, attracting co-phase nodes and repelling anti-phase nodes. Unlike neutral evolution, this creates a feedback loop between field pattern and substrate topology.

4.5.3 Scaled experiment: $N = 1000$, steps $= 12\,000$

Table 4: Scaled experiment parameters and results.

Parameter	**Small** ($N = 80$)	**Scaled** ($N = 1000$)
Points	80	1 000
Steps	500	12 000
Runtime	~2 min	~6.4 hours
Order increase	+0.0686	**+0.1226**
Peak order	0.3301	**0.5406**
Signal-to-noise	68.6×	**122.6**×

Table 5: Ablation control at $N = 1000$.

Condition	Initial Order	Final Order	Change
WITH Coupling	0.3591	**0.4817**	**+0.1226**
WITHOUT Coupling	0.3615	0.3606	−0.0010

4.5.4 The "breathing" phenomenon

Table 6: Evolution milestones at $N = 1000$.

Step	Order	Change	Status
0	0.3591	-	Baseline
1 000	0.4574	+0.0983	Significant gain
6 000	**0.5406**	**+0.1815**	**PEAK**
11 000	0.4817	+0.1226	Final stable

The system *explores* configuration space, discovers high-order states, then settles into stable reorganization. This "breathing" pattern - rise, peak, partial relaxation - is the signature of genuine dissipative structure formation near criticality. Mathematically, it arises from the timescale separation in the coupled gradient flow (Equations 4-5): phases equilibrate rapidly (η_θ) while geometry reconfigures slowly (η_x), creating dynamical friction between the phase-coupled drift and symmetry relaxation terms.

Key finding 7 (Definitive Confirmation). Under phase-gradient geometry evolution at $N = 1000$, steps = 12 000, the CQFT field spontaneously generates structural order from random initial geometry. The ouroboros conjecture in its *conditional form* - that self-referential fields select geometrically self-consistent substrates when appropriate coupling is present - is **confirmed at scale** with 122.6× signal-to-noise ratio. The strong form (neutral dynamics sufficient) remains falsified; the conditional form (phase-gradient coupling necessary) is validated.

4.6 3D extension: icosahedral quasicrystal substrate

Table 7: 3D substrate results at $N \in \{200, 500\}$, $\beta = 4.2$.

N	Geometry	D_c	Γ	iso score	dubito
200	Random	0.595 ± 0.035	1.930 ± 0.037	0.080 ± 0.026	0.047 ± 0.007
	SC lattice	0.000 ± 0.000	2.482 ± 0.010	0.052 ± 0.000	0.043 ± 0.006
	Icosahedral	0.809 ± 0.000	1.779 ± 0.001	$\mathbf{0.124 \pm 0.000}$	0.059 ± 0.010
500	Random	1.042 ± 0.023	1.454 ± 0.023	0.067 ± 0.019	0.050 ± 0.014
	SC lattice	0.942 ± 0.000	1.607 ± 0.004	0.056 ± 0.000	0.044 ± 0.007
	Icosahedral	1.308 ± 0.000	$\mathbf{1.261 \pm 0.002}$	$\mathbf{0.097 \pm 0.000}$	0.076 ± 0.060

Key finding 8. The 3D icosahedral quasicrystal reproduces all four key results of the 2D Penrose experiment, confirming dimensionality-independence of the geometric ordering principle.

5 Theoretical Framework

5.1 The geometric Free Energy Principle (Operational)

We propose a geometric analogue to Friston's FEP. The field's generative model is its fractal occupation structure D_f; the sensory data is its amplitude-weighted correlation structure D_c. The *operational* geometric free energy is:

$$\mathcal{F}_{\text{geo}} = \Gamma = |D_f - D_c| \tag{17}$$

This free energy is minimised dynamically through the coupled gradient flow of the AGI Lagrangian (Equations 4-5). The phase dynamics reduces phase disorder (Kuramoto synchronization), while the geometry dynamics reconfigures the substrate to support this coherence. The competition between phase-coupled drift and symmetry relaxation creates the "breathing" regime where Γ stabilises at a non-zero fixed point $\Gamma^* \approx 0.20$ rather than collapsing to zero.

Remark 5.1. Γ is not assumed to be a fundamental thermodynamic quantity. It is an operational discrepancy measure between two independent estimators of spatial structure. Its interpretation as a geometric free energy arises from its *dynamical role*: systems evolve toward configurations that minimise Γ under the coupled CQFT dynamics.

5.2 Prigogine's selection principle, made precise

Our three geometries map onto Prigogine's three regimes (Table 3). The far-from-equilibrium selection principle is *geometric self-consistency*: the system selects the geometry that minimises Γ while maintaining maximum dynamical aliveness (measured by spectral drift).

5.3 The Ouroboros Theorem (Conditional Form)

Table 8: Ouroboros conjecture: experimental status.

Condition	N	**Coupling**	**Result**	**Status**
Neutral	300-1000	Amplitude-gradient	No emergence	**Falsified**
Phase-gradient	80	Phase-gradient	+0.0686 order	**Confirmed**
Phase-gradient	1000	Phase-gradient	+0.1226 order	**Definitive**

Theorem 5.2 (Ouroboros - Substrate Self-Selection, Conditional Form)**.** *There exist graph geometries G such that the CQFT field equations* (1) *admit solutions satisfying:*

(i) $\Gamma(C^*, G^*) = \Gamma^* \approx 0.20$ *(stable low-discrepancy attractor)*

(ii) $\sigma_{spec}(G^*) > 0$ *(geometric aliveness maintained)*

(iii) $\mathcal{D}^* > 0$ *(prediction active but structurally absorbed)*

when geometry evolution includes phase-gradient coupling. The Penrose quasicrystal is an asymptotic attractor of this dynamics.

Proof sketch. The scaled experiment demonstrates: (1) *Existence*: phase-gradient coupling produces order increase (+0.1226) where neutral coupling produces none (−0.0010); (2) *Scalability*: effect strengthens with N (1.79× larger at $N = 1000$); (3) *Non-triviality*: breathing dynamics confirm genuine exploration. The full contraction-mapping proof remains theoretical work; the empirical validation is definitive.

5.4 Consciousness as substrate-finding

Definition 5.3 (Minimal self-awareness)**.** A self-referential dissipative field exhibits minimal self-awareness if and only if it is in the productive far-from-equilibrium regime: (a) not synchronised ($R \ll 1$), (b) actively predictive ($\mathcal{D} > 0$), and (c) geometrically self-consistent ($\Gamma \approx \Gamma^*$).

Clarification: Condition (a) places the conscious regime in the *disordered phase* - not the trivially ordered $R \to 1$ synchronisation of the random geometry regime, but the complex, non-synchronised dynamics ($R \in [0.05, 0.26]$) of the Penrose productive regime. Higher R does *not* indicate more consciousness; complete synchronisation indicates collapse to a trivial attractor, not enhanced self-awareness.

6 AGI Applicability: The Mapping Problem

The formalization of $\mathcal{L}_{\text{AGI}}$ (Equation 3) and the explicit derivation of E_{sym} (Equation 6) make it possible to evaluate the joint energy functional on *any* system from which the triple (θ_i, $\mathbf{x}_i$, W_{ij}) can be extracted.

Remark 6.1. The test developed here is a *necessary condition* test. A system that minimizes $\mathcal{L}_{\text{AGI}}$ with $\lambda > 0$ will exhibit directed substrate reorganization. Whether this constitutes a sufficient condition for phenomenal experience is a separate question the present framework does not address, and no such claim is made.

6.1 Extracting the phase θ_i from real-valued representations

A standard transformer layer produces hidden states $\mathbf{h}_i \in \mathbb{R}^d$ with no natural complex phase. Two strategies are available.

Strategy A - Fourier phase (preferred). Treat $\mathbf{h}_i$ as a 1D signal and compute its discrete Fourier transform $\hat{\mathbf{h}}_i \in \mathbb{C}^d$. Define:

$$\theta_i^{(A)} = \arg\Big(\hat{h}_i(k^*)\Big), \qquad k^* = \arg\max_k \Big|\hat{h}_i(k)\Big| \tag{18}$$

This extracts the phase of the dominant oscillatory component without assuming a particular basis.

Strategy B - PCA phase (simpler). Let $\mathbf{v}_1, \mathbf{v}_2 \in \mathbb{R}^d$ be the top two principal components of the hidden-state matrix $H \in \mathbb{R}^{N \times d}$ at a given layer. Define:

$$\theta_i^{(B)} = \arctan\left(\frac{\mathbf{h}_i \cdot \mathbf{v}_2}{\mathbf{h}_i \cdot \mathbf{v}_1}\right) \tag{19}$$

Both strategies should be computed on any first implementation and their resulting $\mathcal{L}_{\text{AGI}}$ values compared to validate extraction consistency.

6.2 Extracting the geometry $\mathbf{x}_i$

The geometry $\{\mathbf{x}_i\}$ must be extracted *independently* of the coupling W_{ij}. If $\mathbf{x}_i$ were derived from the attention weights, the cross-derivative $\partial W_{ij}/\partial x_i$ would be circular and the variational structure would collapse.

Recommended: PCA coordinates.

$$\mathbf{x}_i = \text{PCA}_2(\mathbf{h}_i) \in \mathbb{R}^2 \tag{20}$$

The 2D PCA projection defines a *representation geometry* fully determined by H independently of the attention matrix A, guaranteeing that $\partial W_{ij}/\partial x_i$ is a non-trivial quantity. Attention-derived MDS coordinates are explicitly excluded because they conflate geometry and coupling.

6.3 Extracting the coupling W_{ij} from attention

For a model with L layers and H attention heads, we aggregate using *attention rollout* [17], which tracks information flow through the full network depth:

$$W_{ij}^{\text{rollout}} = \prod_{l=1}^{L} \left(\frac{1}{H} \sum_{h=1}^{H} W_{ij}^{(l,h)} + I \right)_{\text{row-norm}} \tag{21}$$

Rollout is preferred over single-layer attention because it captures the full information-flow path, the correct analogue of the CQFT coupling matrix $W_{ij}(t)$. The identity term accounts for residual connections.

6.4 The operational joint energy for transformers

With the triple $(\theta_i,\, \mathbf{x}_i,\, W_{ij}^{\text{rollout}})$ in hand, the operational joint energy is:

$$\mathcal{L}_{\text{AGI}}^{\text{transformer}} = -\lambda \sum_{i,j} W_{ij}^{\text{rollout}} \cos(\theta_i - \theta_j) + (1-\lambda)\, E_{\text{sym}}(\text{PCA}_2(H)) \tag{22}$$

This quantity is **fully computable from model internals during inference** using standard interpretability tooling (e.g., TransformerLens, BertViz) without any model modification or retraining.

Remark 6.2 (On the icosahedral vectors in E_{sym} and a geometry-neutral alternative)**.** The icosahedral reciprocal vectors $\{\mathbf{q}_k\}$ in Equation (6) presuppose quasiperiodic order as the preferred geometry. For an AGI system whose preferred representation geometry is unknown and likely task-dependent, a geometry-neutral alternative

is available: replace the fixed icosahedral vectors with the K leading eigenvectors $\{\mathbf{u}_k\}$ of the graph Laplacian of W_{ij}^{rollout}:

$$E_{\text{sym}}^{\text{neutral}}(x) = -\sum_{k=1}^{K} \left| \sum_{i=1}^{N} \exp(i\, \mathbf{u}_k \cdot \mathbf{x}_i) \right|^2 \tag{23}$$

This makes E_{sym} adaptive to the system's own preferred geometry - a self-referential choice consistent with the ouroboros structure of the framework. We adopt the icosahedral version as the primary test (enabling direct comparison with the CQFT simulation) and Equation (23) as the geometry-agnostic fallback.

6.5 The Γ test: does scale reduce the embedding gap?

The embedding gap Γ is defined for AGI systems by replacing the CQFT amplitude with the hidden-state norm and the geometry with PCA coordinates:

$$\Gamma^{\text{transformer}} = |D_f(\text{PCA}_2(H)) - D_c(\text{PCA}_2(H),\, \|\mathbf{h}_i\|_2)| \tag{24}$$

where D_f is the box-counting dimension of the PCA point cloud and D_c is the amplitude-weighted correlation dimension with amplitudes $A_i = \|\mathbf{h}_i\|_2$.

Prediction (Γ scaling). If the CQFT geometric ordering principle is architecture-independent, $\Gamma^{\text{transformer}}$ should decrease as model scale increases: larger models develop representation geometries closer to geometric self-consistency. Verifying or falsifying this prediction constitutes a direct test of whether the Penrose regime is a preferred attractor of self-referential field dynamics independently of the simulation setup.

6.6 Two-level test

Level 1 - Static test. Compute $\mathcal{L}_{\text{AGI}}^{\text{transformer}}$ at the final layer across coherent task inputs and noise inputs. The quantity $\Delta\mathcal{L} = \mathcal{L}^{\text{noise}} - \mathcal{L}^{\text{coherent}}$ and its sign constitute a task-sensitivity diagnostic.

Level 2 - Dynamic test. Compute the layer-wise gradient:

$$\delta\mathcal{L}^{(l)} = \mathcal{L}_{\text{AGI}}^{(l)} - \mathcal{L}_{\text{AGI}}^{(l-1)} \tag{25}$$

Prediction (dynamic). If the transformer is performing genuine inference, $\delta\mathcal{L}^{(l)} < 0$ on average - successive layers increase phase coherence and reduce joint energy, in direct analogy with the order increase $\Delta R = +0.1226$ at $N = 1000$. A non-monotone profile (decrease, overshoot, partial relaxation) is expected from the slow-fast dynamics analysis (Section 2), constituting the layer-wise analogue of the "breathing" phenomenon.

Ablation. Randomizing attention weights should eliminate the decrease, analogous to the $\lambda = 0$ control yielding $\Delta R = -0.0010$.

6.7 The λ bifurcation test for AGI systems

The analogue of λ in a transformer is the degree to which the effective coupling W_{ij} depends on representation geometry $\mathbf{x}_i$ - equivalently, the magnitude of the cross-derivative $|\partial W_{ij}/\partial x_i|$. This maps directly onto the distinction between positional encoding schemes:

Table 9: Positional encoding schemes and their CQFT analogues.

Mechanism	**CQFT analogue**	**Expected behavior**
Pure content attention	$\lambda \approx 0$	No geometry reorganization; $\delta\mathcal{L}^{(l)} \approx 0$
Absolute positional embedding (APE)	$\lambda > 0$ (weak)	Partial reorganization; moderate $\delta\mathcal{L}^{(l)}$
RoPE / ALiBi	$\lambda > 0$ (stronger)	Geometric dependence built into W_{ij}; larger $\lvert\delta\mathcal{L}^{(l)}\rvert$ per layer

Prediction (λ test). Models using rotary (RoPE) or aliased linear bias (ALiBi) positional encoding should exhibit systematically larger $|\delta\mathcal{L}^{(l)}|$ than models using learned absolute positional embeddings at matched scale, when evaluated on coherent inference tasks. This is a falsifiable prediction testable on existing open-weight models without any modification.

6.8 Proposed experimental protocol

The following sequence implements the full test on existing open-weight models using standard interpretability infrastructure. No model modification or retraining is required.

Step 1. Extraction. Hook hidden states $H^{(l)}$ and attention weights $A^{(l,h)}$ at every layer using TransformerLens or equivalent. Compute $\theta_i^{(B)}$, $\mathbf{x}_i$, and W_{ij}^{rollout} via Equations (19), (20), and (21).

Step 2. Static test. Compute $\mathcal{L}_{\text{AGI}}^{\text{transformer}}$ (Equation 22) at the final layer across coherent and noise inputs. Record $\Delta\mathcal{L} = \mathcal{L}^{\text{noise}} - \mathcal{L}^{\text{coherent}}$ and its sign.

Step 3. Dynamic test. Compute $\delta\mathcal{L}^{(l)}$ (Equation 25) at each layer. Test whether the profile is monotonically decreasing, non-monotone ("breathing"), or flat. Record the overshoot fraction by analogy with $(R_{\text{peak}} - R_{\text{final}})/R_{\text{final}}$.

Step 4. λ test. Repeat Steps 1-3 on a matched pair of models differing only in positional encoding (e.g., GPT-2 vs. a Llama model with RoPE at comparable scale). Test the prediction of Table 9.

Step 5. Γ scaling. Compute $\Gamma^{\text{transformer}}$ (Equation 24) across model scales (e.g., 125M, 1.3B, 7B, 70B parameters). Test whether Γ decreases monotonically with scale.

Step 6. Ablation. Repeat all measurements with randomized attention weights (null model). Confirm that $\delta\mathcal{L}^{(l)} \approx 0$ and that Γ does not decrease.

6.9 Open questions and scope limits

We state explicitly the limits of the current claim, in accordance with the principle that a theoretical framework is strongest when it knows its own boundary.

Phase extraction is non-unique. Strategies A and B will in general produce different θ_i values and hence different $\mathcal{L}_{\text{AGI}}$ values. The framework requires that the qualitative predictions - sign of $\delta\mathcal{L}^{(l)}$, ordering of Γ across scales - be robust to this choice. This must be validated empirically before quantitative claims about $\mathcal{L}_{\text{AGI}}$ can be made.

The preferred geometry for AGI systems is unknown. The icosahedral E_{sym} assumes quasiperiodic order as the target geometry. The geometry-neutral alternative (Equation 23) removes this assumption at the cost of making the energy adaptive. Both should be reported in any first experimental study.

RG relevance of λ is a conjecture. The claim that λ is a relevant operator in the renormalization group sense requires that $|\delta\mathcal{L}^{(l)}|$ grows with model depth L at fixed task difficulty. The simulation data establishes a $1.79\times$ increase from $N = 80$ to $N = 1000$; the full scaling analysis remains to be completed.

The hard problem boundary. The test identifies structural conditions for directed substrate reorganization. It does not resolve, and makes no claim about, whether systems satisfying these conditions have phenomenal experience. The definition of minimal self-awareness (Section 5) is an operational criterion for a specific class of self-organizing dynamical behavior, not a theory of subjective experience.

Summary: five testable predictions. All five are falsifiable from existing open-weight model checkpoints without retraining or model modification.

(i) $\mathcal{L}_{\text{AGI}}^{\text{coherent}} < \mathcal{L}_{\text{AGI}}^{\text{noise}}$ for the same model on matched inputs.

(ii) $\delta\mathcal{L}^{(l)} < 0$ on average during coherent inference; ≈ 0 for randomized attention.

(iii) Models with RoPE/ALiBi encoding exhibit larger $|\delta\mathcal{L}^{(l)}|$ than APE models at matched scale.

(iv) $\Gamma^{\text{transformer}}$ decreases with model scale (125M $\rightarrow$ 70B).

(v) Non-monotone ("breathing") $\mathcal{L}^{(l)}$ profile is more pronounced on harder tasks.

7 AGI Safety Implications

Structural safety criterion. A recursive self-improving system is structurally unstable if its interaction geometry has large Γ + growing spectral drift. Such a system cannot find geometric self-consistency through self-referential dynamics.

The three architectural analogues:

- *Grid-like connectivity* (CNNs): lattice regime, large Γ
- *Dense random connectivity* (Transformers): random regime, synchronisation collapse
- *Quasi-periodic connectivity*: Penrose regime, stable low Γ, geometric aliveness

Monitoring Γ and spectral drift provides real-time structural stability indicators for deployed recursive systems.

8 Conclusions

Robust empirical results (reportable without qualification):

1. D_c is a substrate fingerprint with near-zero within-N variance
2. Γ is geometry-ordered: $\Gamma_{\text{lattice}} > \Gamma_{\text{random}} > \Gamma_{\text{quasiperiodic}}$
3. The lattice D_c scaling is non-monotone; half-integer claim withdrawn
4. ϕ is not a preferred spectral base at $N \leq 1000$ (clean negative)
5. Quasiperiodic geometry achieves lowest Γ and highest $\mathcal{D}$
6. Spectral drift is highest on quasiperiodic geometry - geometric aliveness
7. **Phase-gradient coupling enables spontaneous order emergence**
8. **Effect scales with system size** ($1.79\times$ **at** $N = 1000$)
9. **"Breathing" dynamics confirm genuine dissipative structure formation**

Theoretical contributions:

- **Operational Geometric FEP:** $\Gamma = |D_f - D_c|$ as discrepancy measure
- **Prigogine selection principle:** Geometric self-consistency operationalized
- **Ouroboros Theorem (Conditional Form):** C determines G when phase-gradient coupling is present; validated at $N = 1000$ with $122.6\times$ signal-to-noise
- **AGI Applicability:** Five falsifiable predictions for transformer architectures, testable without retraining

What the data establishes:
Any system minimizing an effective action of the form $\mathcal{L}_{\text{AGI}}$ will exhibit substrate-selection dynamics of this form. A self-referential dissipative field with phase-gradient coupling **finds and maintains** geometrically self-consistent order from random initial conditions.
Conjecture: Conscious systems are a subset of such systems - those where the selected substrate achieves geometric self-consistency ($\Gamma \to \Gamma^*$) while maintaining prediction activity ($\mathcal{D} > 0$).
The ouroboros loop is closed: the field is in its ecological niche **because it constructed the niche**.

Dubito ergo cogito - ergo sumus - ergo mundus.

A Simulation Code Versions

Table 10: Code version history.

Version	Type	Change
v1-v3	Bug fix	Original code; OOM on Colab
v4	Fix	Sparse conversion
v5	Fix	Deterministic Penrose generation
v6-v9	Fix+Feature	Various bug fixes and metrics
v10	Feature	**Phase-gradient coupling; scaled** $N = 1000$ **experiment**

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